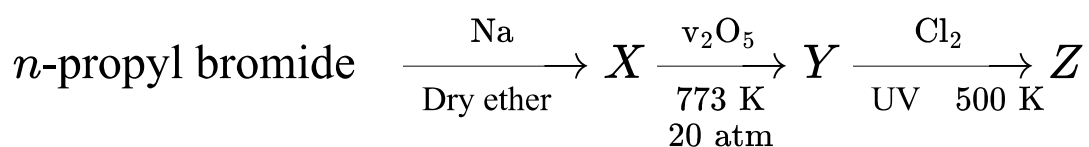


Hydrocarbons

Question1

The empirical formula weight of ' Z ' in the given reaction sequence is



AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

47.5

B.

54.5

C.

84.5

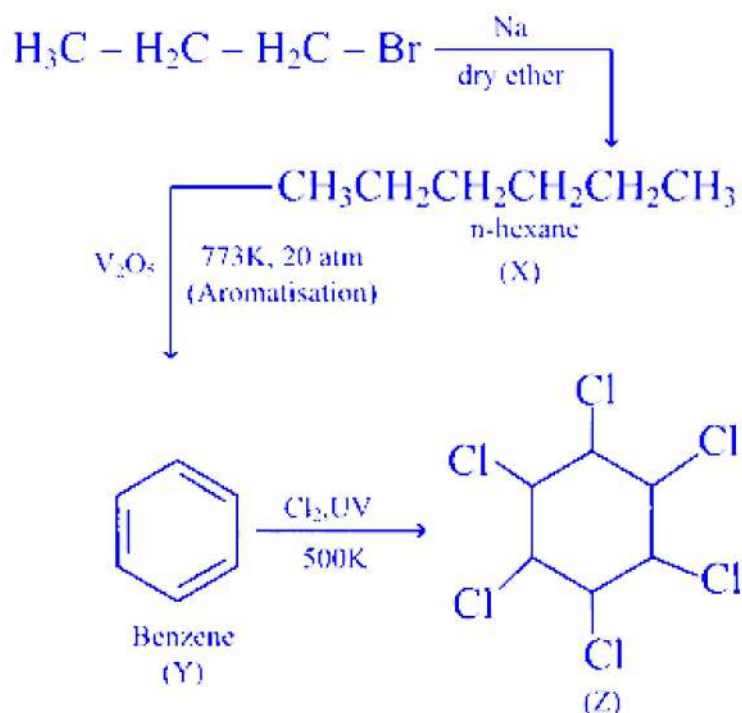
D.

48.5

Answer: D

Solution:

The complete reaction sequence is as follows



So, the molecular formula is $\text{C}_6\text{H}_6\text{Cl}_6$ and its empirical formula is CHCl . Thus, the empirical formula weight of Z is 48.5

Question2

Which one of the following compounds does not give benzoic acid when treated with alkaline KMnO_4 ?

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A. Acetophenone

B.

n-propyl benzene

C.

Styrene

D.

t-butyl benzene

Answer: D

Solution:

t-butyl benzene does not give benzoic acid when treated with alkaline KMnO_4 because it lacks the necessary benzylic hydrogen required for oxidation to form benzoic acid, since the tertiary carbon in *t*-butylbenzene has no hydrogen attached directly to it, the oxidation cannot proceed to form carboxylic acid.

Question3

Consider the following

Statement I Kolbe's electrolysis of sodium propionate gives *n*-hexane as product.

Statement II In Kolbe's process CO_2 is liberated at anode and H_2 is liberated at cathode.

Correct answer is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

Both statement-I and statement-II are correct

B.

Both statement-I and statement-II are not correct

C.

Statement-I is correct, but statement-II is not correct

D.

Statement-I is not correct, but statement-II is correct

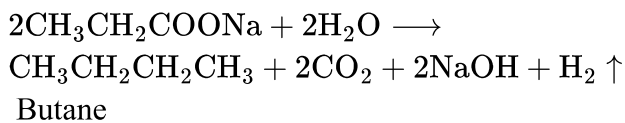
Answer: D

Solution:

Statement I is wrong, but Statement II is right.

Statement I should say: In Kolbe's electrolysis of sodium propionate, the product is butane, not n-hexane.

The chemical reaction is:



In Statement II, it is correct that in Kolbe's process, carbon dioxide (CO_2) is released at the anode and hydrogen gas (H_2) is released at the cathode.

Question4

The catalyst used for the isomerisation of *n*-alkanes to branched chain alkanes is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

anhy. AlCl_3/HCl

B.

Mo_2O_3

C.

FeCl_3

D.

$\text{TiCl}_4 + \text{R}_3\text{Al}$

Answer: A

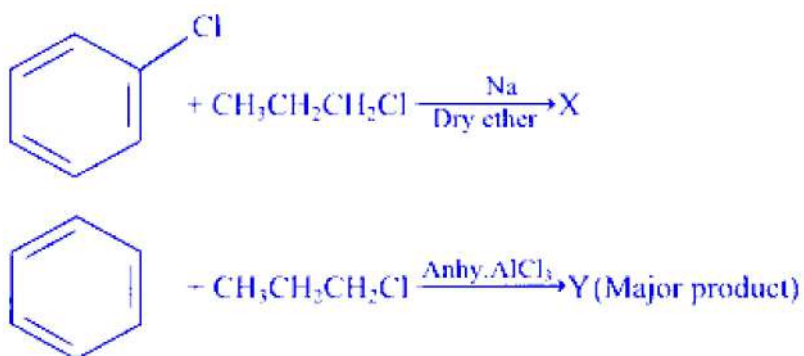
Solution:

The primary catalyst used for the isomerisation of *n*-alkanes to branched chain alkanes are acid catalyst, particularly zeolites and sulphated zirconia.

Anhydrous AlCl_3 or HCl can be also used.

Question5

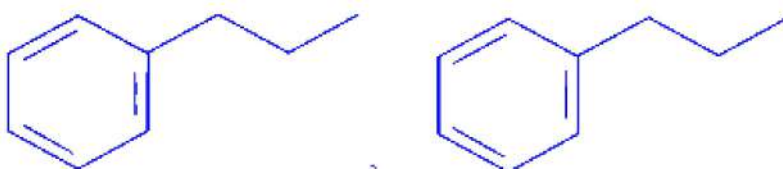
What are X and Y respectively, in the following set of reactions?



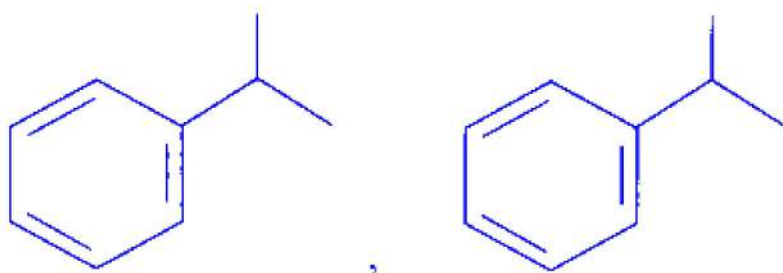
AP EAPCET 2025 - 21st May Evening Shift

Options:

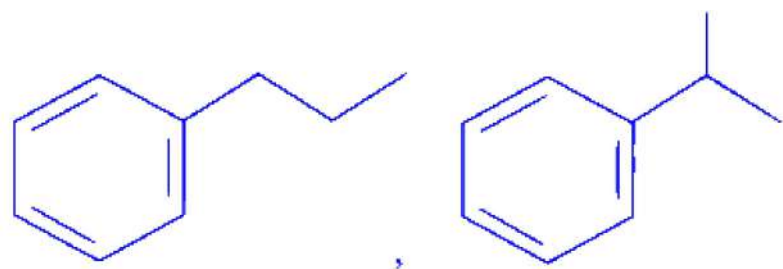
A.



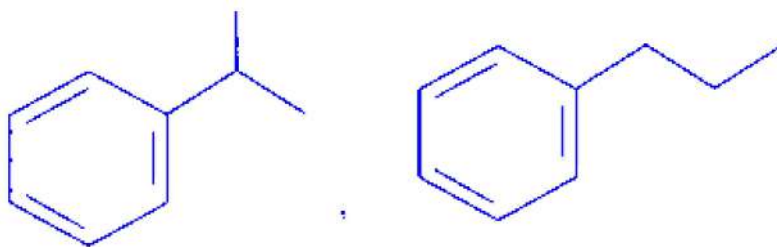
B.



C.



D.

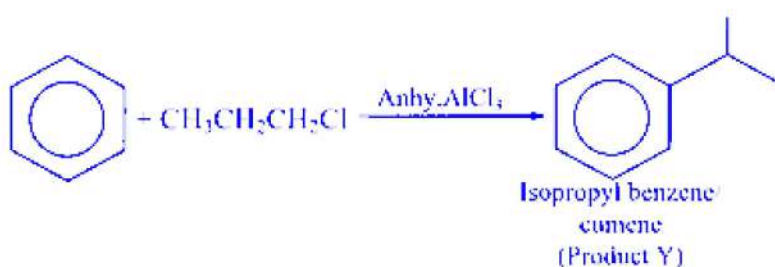


Answer: C

Solution:

The complete reaction is as follows

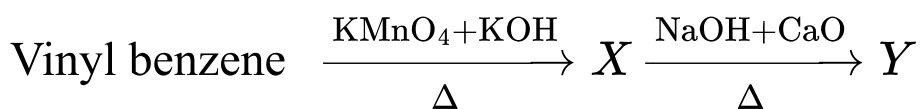
This is Wurtz-fittig reaction



This is Friedel-Craft alkylation.

Question6

Consider the following reaction sequence



'Y ' can also be formed from

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

polymerisation of ethylene

B.

polymerisation of propyne

C.

aromatisation of *n*-hexane

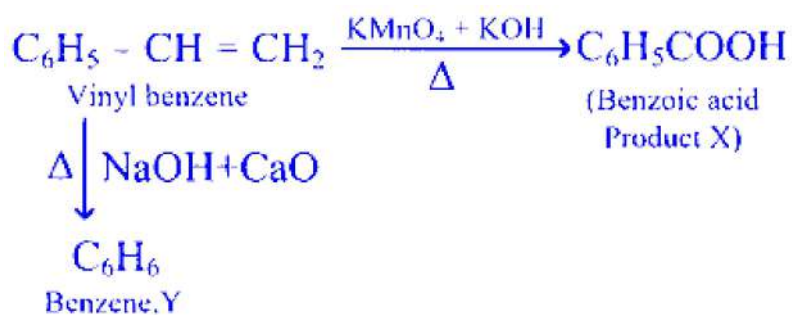
D.

aromatisation of *n*-heptane

Answer: C

Solution:

The complete reaction is as follows,



The final product is benzene, which is also formed by aromatisation of *n*-hexane.

Question7

When ethyl bromide and *n*-propyl bromide are allowed to react with Na metal in dry ether, the number of different alkanes formed is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

1

B.

2

C.

3

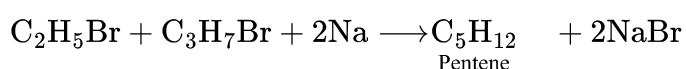
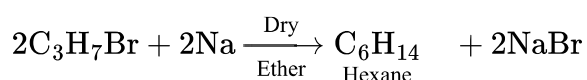
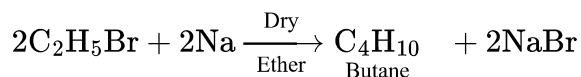
D.

4

Answer: C

Solution:

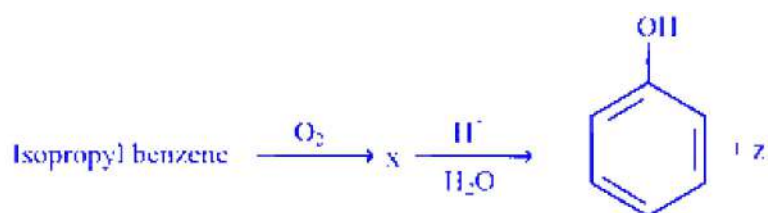
The reaction is a Wurtz reaction, In the given case, three alkanes are formed



∴ Butane, hexane, pentane are formed.

Question8

Consider the following sequence of reactions



The incorrect statement about z is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

z gives yellow precipitate of CHI_3 with $\text{NaOH} + \text{I}_2$, solution.

B.

z gives isopropyl alcohol on reduction with H_2 in the presence of Pd catalyst.

C.

z on reaction with CH_3MgBr followed by hydrolysis gives 2° alcohols.

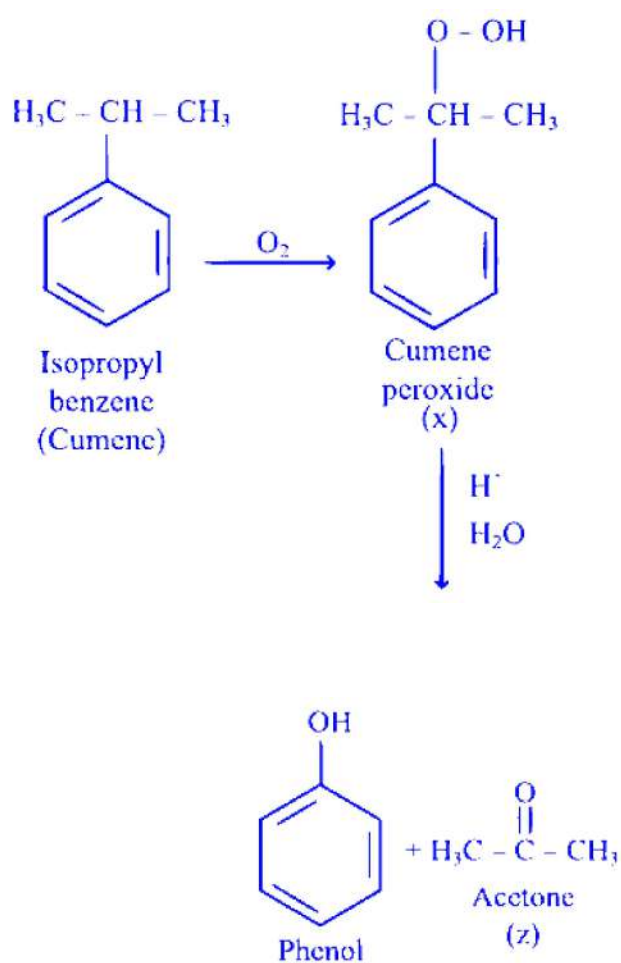
D.

z does not give positive test with Fehling's reagent.

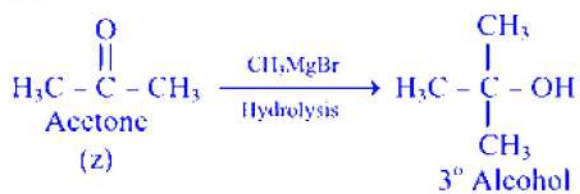
Answer: C

Solution:

The complete reaction sequence is as follows



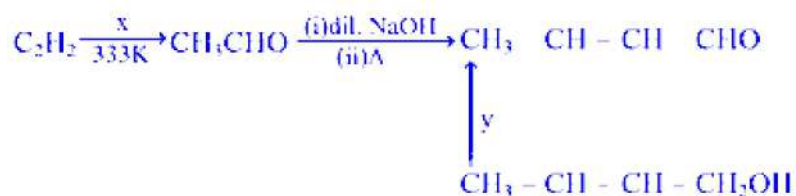
Now



Thus, statement given in option (C) is incorrect as on reaction with CH_3MgBr followed by hydrolysis will give 3° alcohol.

Question9

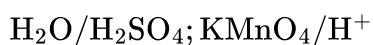
What are x and y in the following reaction sequence?



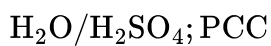
AP EAPCET 2025 - 21st May Morning Shift

Options:

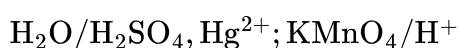
A.



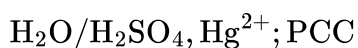
B.



C.



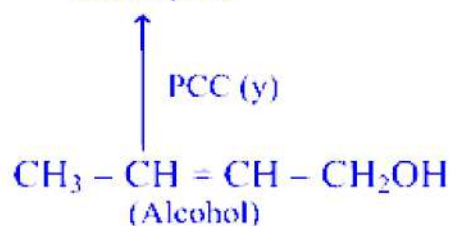
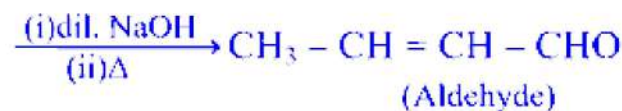
D.



Answer: D

Solution:

The complete reaction is as follows,



$x = \text{H}_2\text{O}/\text{H}_2\text{SO}_4, \text{Hg}^{2+}$

$y = \text{PCC}$

Question10

Which of the following sequence of reagents convert 3- hexene to propane?

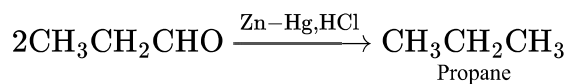
AP EAPCET 2024 - 23th May Morning Shift

Options:

- A. $\text{KMnO}_4\text{H}^+, \text{NaOH}, \text{CaO}$
- B. (i) O_3 (ii) $\text{Zn}, \text{H}_2\text{O}, \text{NaBH}_4$
- C. (i) O_3 (ii) $\text{Zn}, \text{H}_2\text{O}, \text{Zn} - \text{Hg}, \text{HCl}$
- D. $\text{KMnO}_4\text{H}^+, \text{LiAlH}_4, \text{H}_2\text{O}$

Answer: C

Solution:



So, the correct sequence of reagent is (i) O_3 (ii) $\text{Zn}, \text{H}_2\text{O}; \text{Zn} - \text{Hg}, \text{HCl}$.

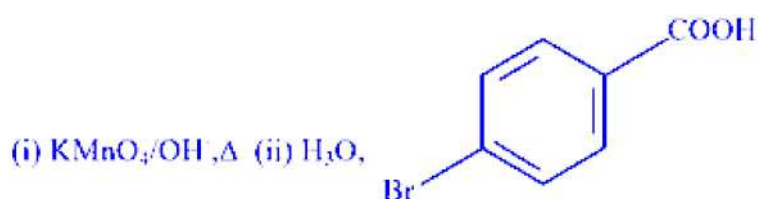
Question11

Toluene on reaction with the reagent X gave Y , which dissolves in NaHCO_3 and when reacted with Br_2/Fe gave Z . What are X and Z ?

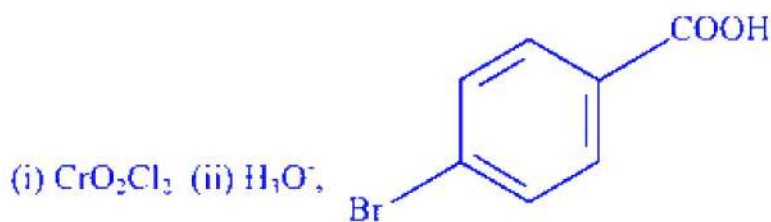
AP EAPCET 2024 - 23th May Morning Shift

Options:

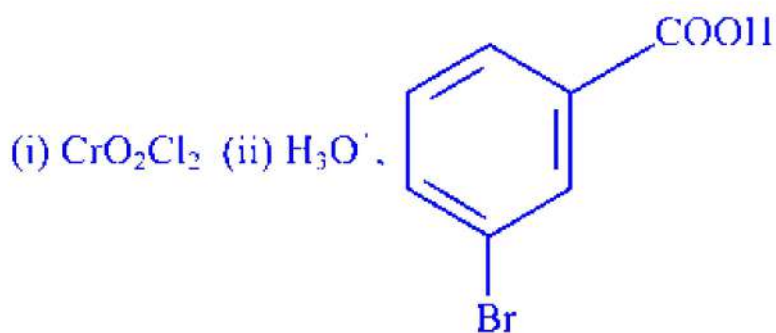
A.



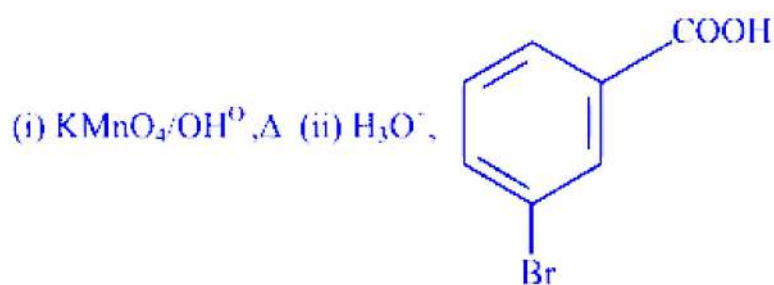
B.



C.

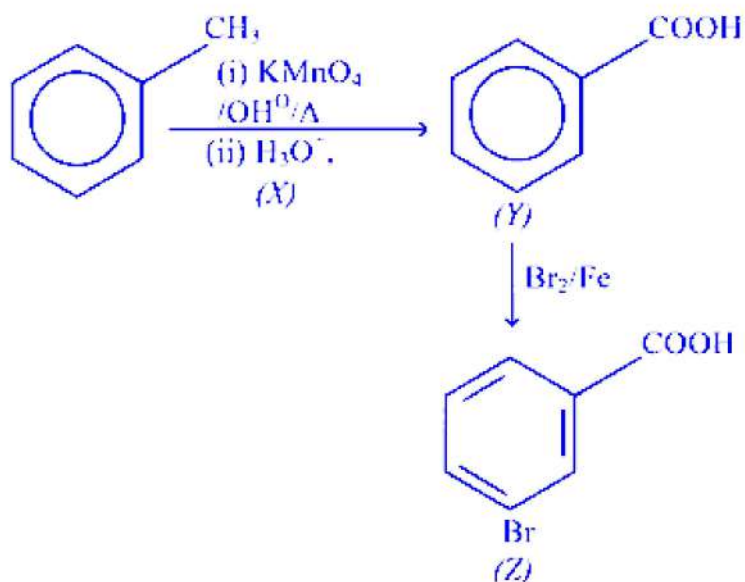


D.



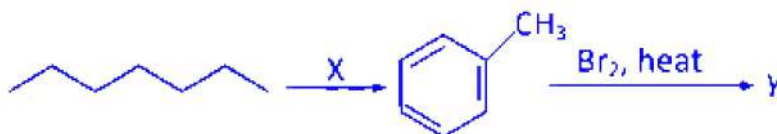
Answer: D

Solution:



Question12

What are X and Y respectively in the following reaction sequence?



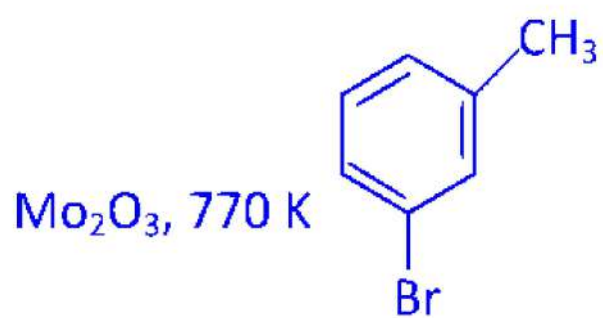
AP EAPCET 2024 - 22th May Evening Shift

Options:

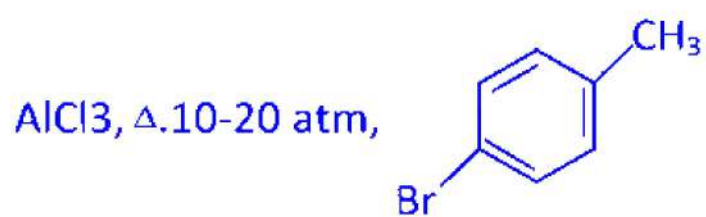
A.



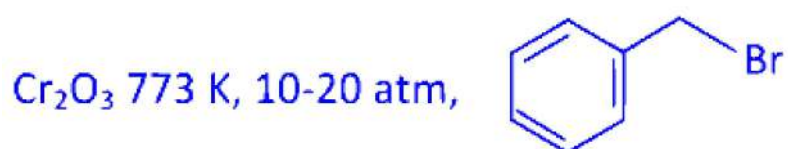
B.



C.



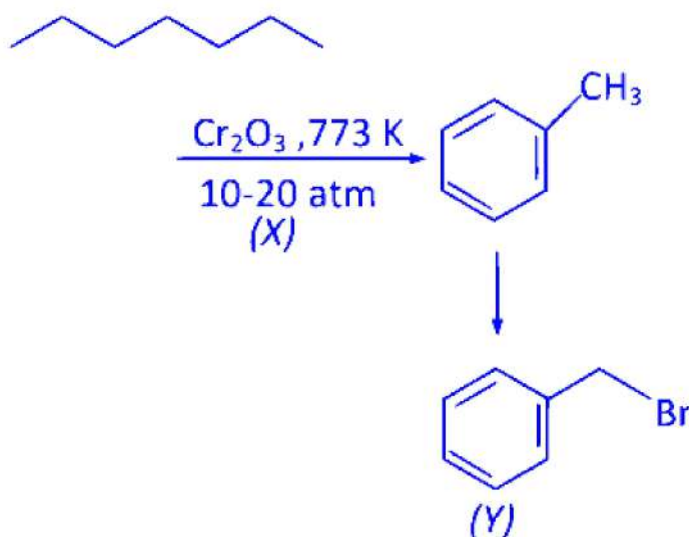
D.



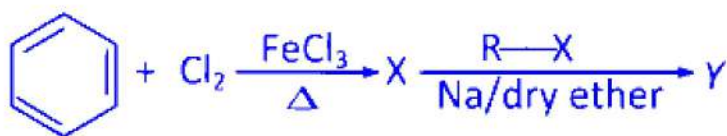
Answer: D

Solution:

The complete reaction is as follows



Question13



Conversion of X to Y is an example of

AP EAPCET 2024 - 22th May Evening Shift

Options:

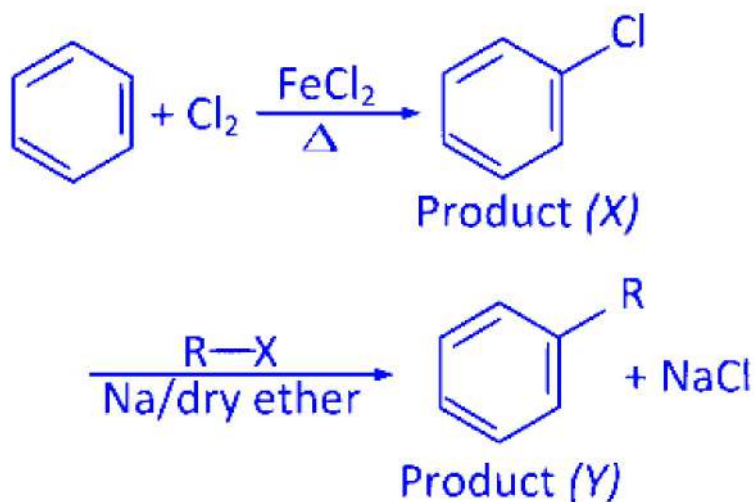
- A. Wurtz reaction
- B. Fittig reaction
- C. Wurtz-Fittig reaction
- D. Friedel-Crafts reaction

Answer: C

Solution:

The given reaction is Wurtz Fitting reaction.

It is a chemical process that produces substituted aromatic compounds from aryl halides and alkyl halides in presence of sodium and dry ether. The complete reactions are as follow,



Question14

In Kolbe's electrolysis of sodium propanoate, products formed at anode and cathode are respectively

AP EAPCET 2024 - 22th May Morning Shift

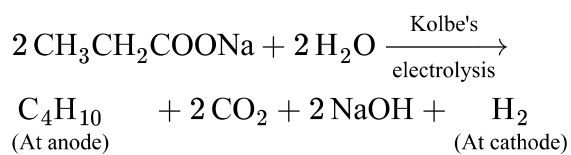
Options:

- A. C₂H₆, H₂
- B. C₃H₈, H₂
- C. C₁H₄, H₂
- D. H₂, C₄H₁₀

Answer: C

Solution:

In Kolbe's electrolysis, the sodium or potassium salt of a carboxylic acid undergoes decarboxylation to form the corresponding alkanes. The overall reaction is outlined below:



Thus, the products at the anode and cathode are:

Anode: C_4H_{10} (butane)

Cathode: H_2 (hydrogen gas)

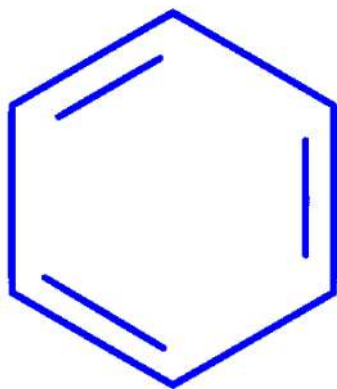
Question15

In Wurtz-Fittig reaction a compound X reacts with alkyl halide. What is X ?

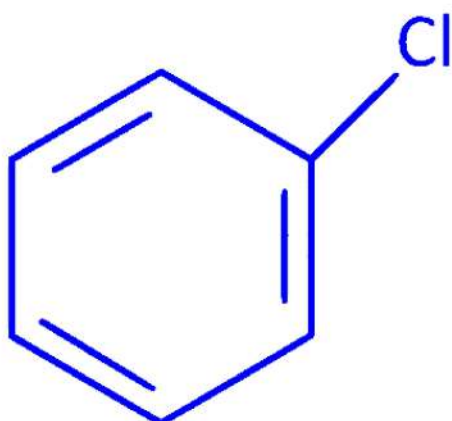
AP EAPCET 2024 - 22th May Morning Shift

Options:

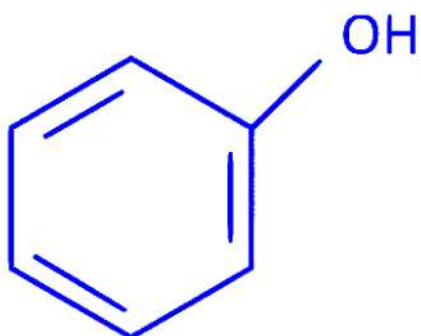
A.



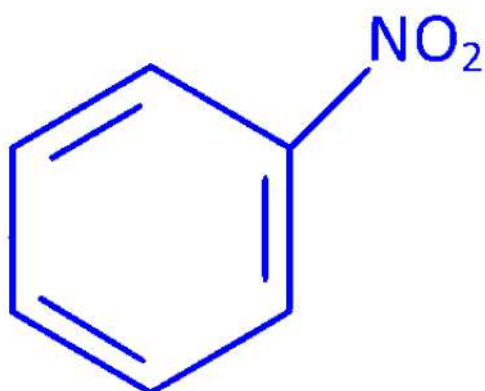
B.



C.



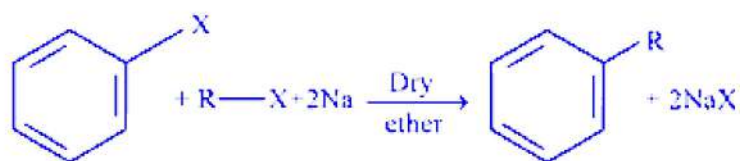
D.



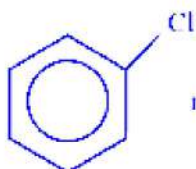
Answer: B

Solution:

Wurtz-Fittig reaction Aryl halide and alkyl halide react in presence of sodium metal/dry ether to form alkyl benzene.



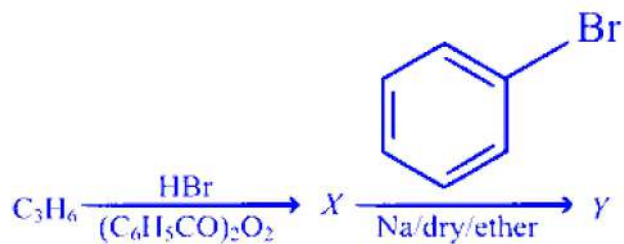
In given option,



match correctly as aryl halides.

Question16

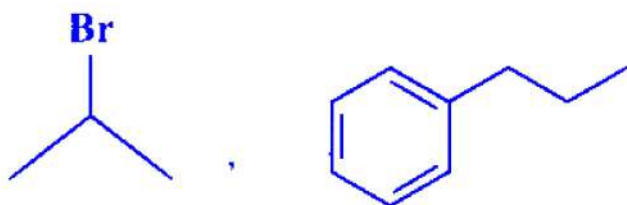
What are X and Y respectively in the following reaction sequence?



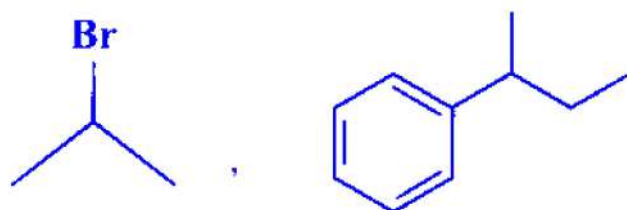
AP EAPCET 2024 - 21th May Evening Shift

Options:

A.



B.



C.



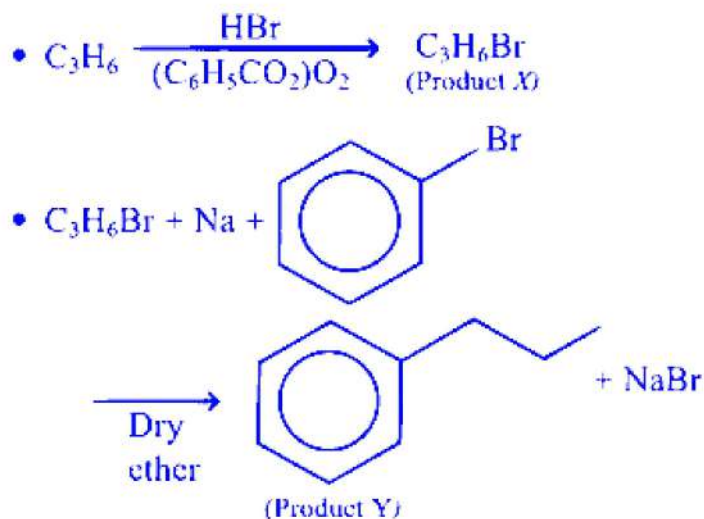
D.



Answer: D

Solution:

The complete reaction is as follows,



This reaction is called as Wurtz reaction.

Question 17

Hydrolysis of an alkyl bromide ($\text{C}_5\text{H}_{11}\text{Br}$) follows first order kinetics. Reaction of X with Mg in dry ether followed by treatment of D_2O gave Y, What is Y ?

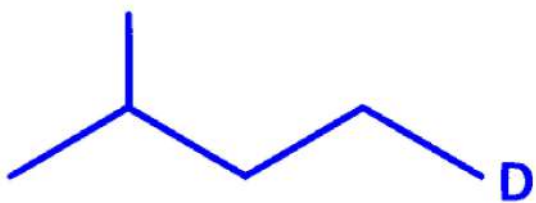
AP EAPCET 2024 - 21th May Evening Shift

Options:

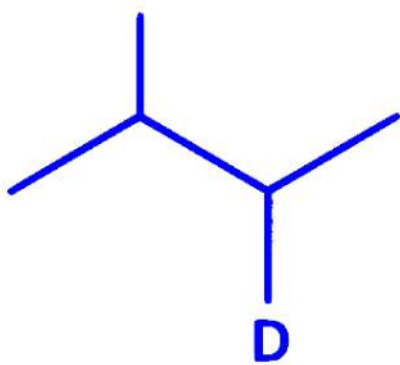
A.



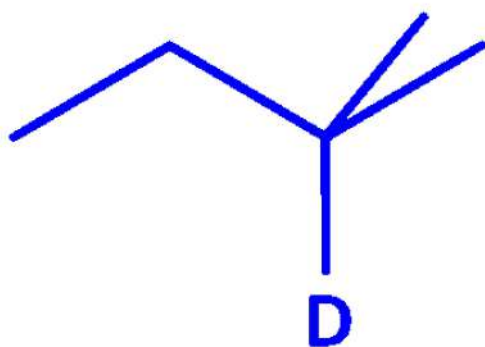
B.



C.

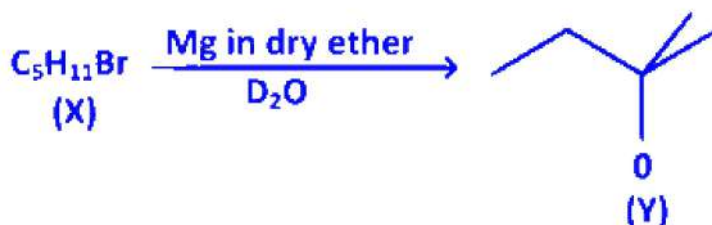


D.



Answer: D

Solution:



Question18

A mixture of ethyl iodide and *n*-propyl iodide is subjected to Wurtz reaction. The hydrocarbon which will not be formed is

AP EAPCET 2024 - 21th May Morning Shift

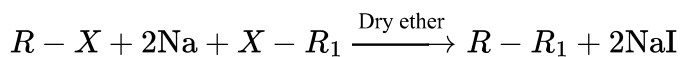
Options:

- A. butane
- B. propane
- C. pentane
- D. hexane

Answer: B

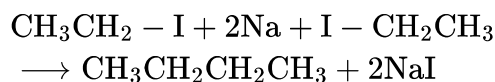
Solution:

In the Wurtz reaction, alkanes are synthesized by reducing alkyl halides using sodium in dry ether. This process combines two alkyl groups to form higher alkanes.

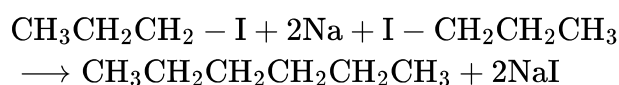


When a mixture of ethyl iodide and *n*-propyl iodide undergoes the reaction, the following higher alkanes are formed:

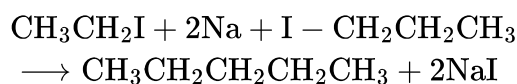
(i) **Butane:**



(ii) **Hexane:**



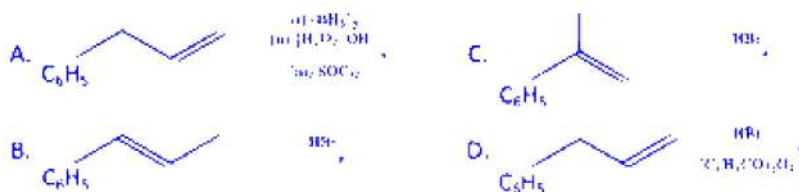
(iii) **Pentane:**



Thus, propane is the only hydrocarbon not formed in this reaction.

Question19

The product of which of the following reactions undergo hydrolysis by $\text{S}_{\text{N}}1$ mechanism?



AP EAPCET 2024 - 20th May Evening Shift

Options:

A. C, D only

B. A, B, C only

C. B, C only

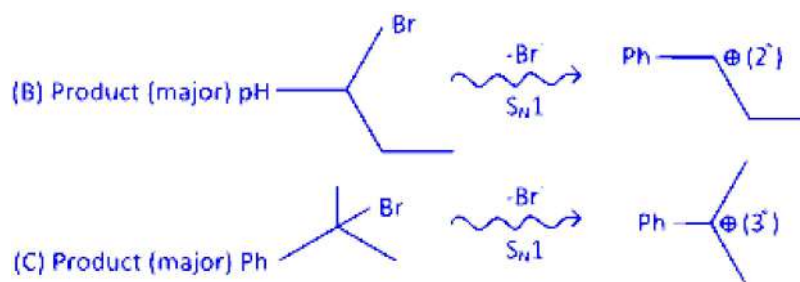
D. A, D only

Answer: C

Solution:

S_N1 mechanism are generally carried out in polar protic solvents and generally occurs in two steps.

Among the given reactions products of *B* and *C* will undergo hydrolysis by S_N1 mechanism, as the products are $2^\circ/3^\circ$ benzy bromides which will furnish very stable carbocations.



Question20

Styrene on reaction with reagent *X* gave *Y* which on hydrolysis followed by oxidation gave *Z*. *Z* gives positive 2, 4-DNP test but does not give iodoform test. What are *X* and *Z* respectively ?

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $\text{HBr} : \text{C}_6\text{H}_5\text{COCH}_3$

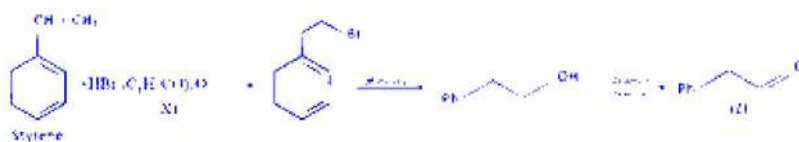
B. $\text{HBr} : \text{C}_6\text{H}_5\text{CH}_2\text{CHO}$

C. $\text{HBr}/(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2 : \text{C}_6\text{H}_5\text{CH}_2\text{CHO}$

D. $\text{HBr}/(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2 : \text{C}_6\text{H}_5\text{COCH}_3$

Answer: C

Solution:



Thus, the compound X and Z are $\text{HBr}/(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$ and $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$.

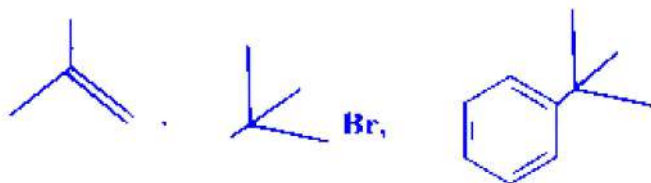
Question21

An alkene X (C_4H_8) on reaction with HBr gave Y ($\text{C}_4\text{H}_9\text{Br}$). Reaction of Y with benzene in the presence of anhydrous AlCl_3 gave Z which is resistant to oxidation with $\text{KMnO}_4 - \text{KOH}$. What are X, Y, Z respectively?

AP EAPCET 2024 - 20th May Morning Shift

Options:

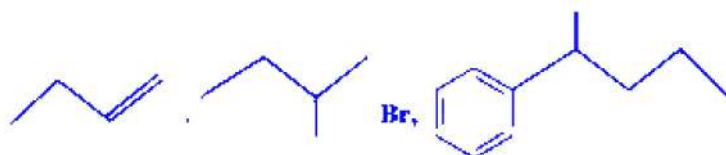
A.



B.



C.



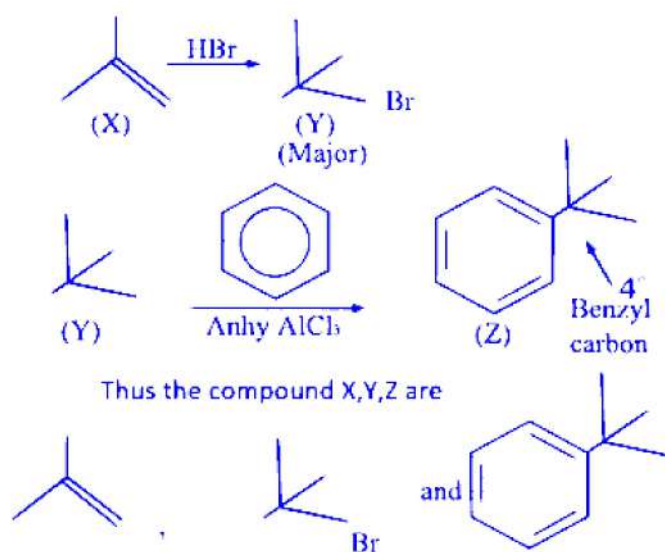
D.



Answer: A

Solution:

The complete reaction is as follows



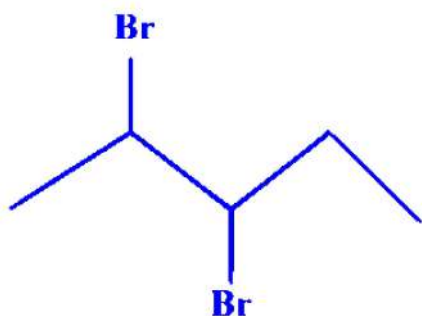
Question22

An alkene X (C_4H_8) does not exhibit cis-trans isomerism. Reaction of X with Br_2 in the presence of UV light gave Y . What is Y ?

AP EAPCET 2024 - 20th May Morning Shift

Options:

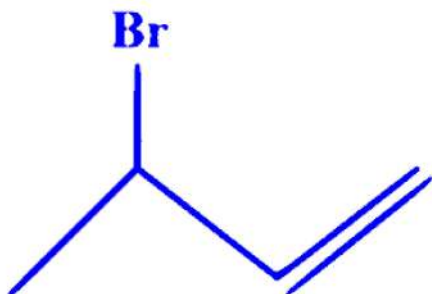
A.



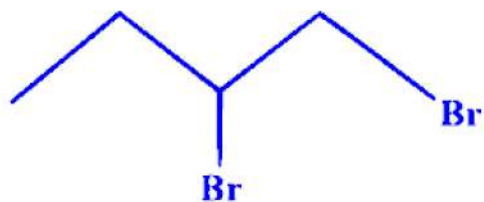
B.



C.



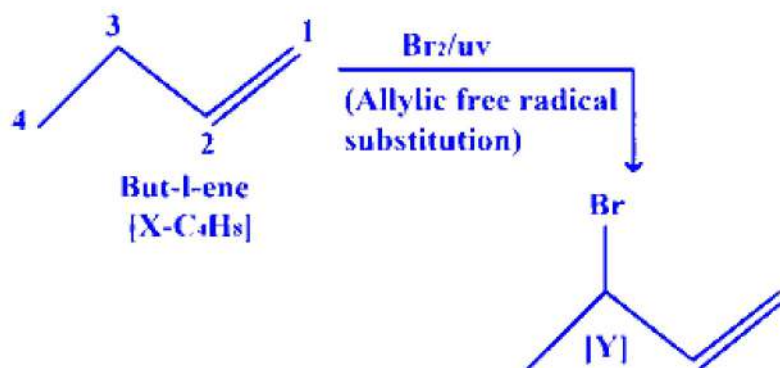
D.



Answer: C

Solution:

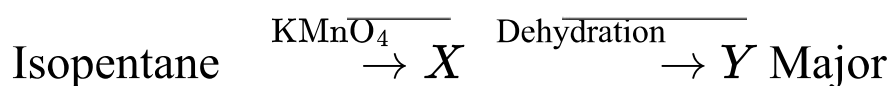
But-1-ene does not show cis-trans isomerism (geometrical).



Here, c-3 of X is the allyl carbon (2°).

Question23

What are X and Y respectively in the following reaction sequence?



AP EAPCET 2024 - 19th May Evening Shift

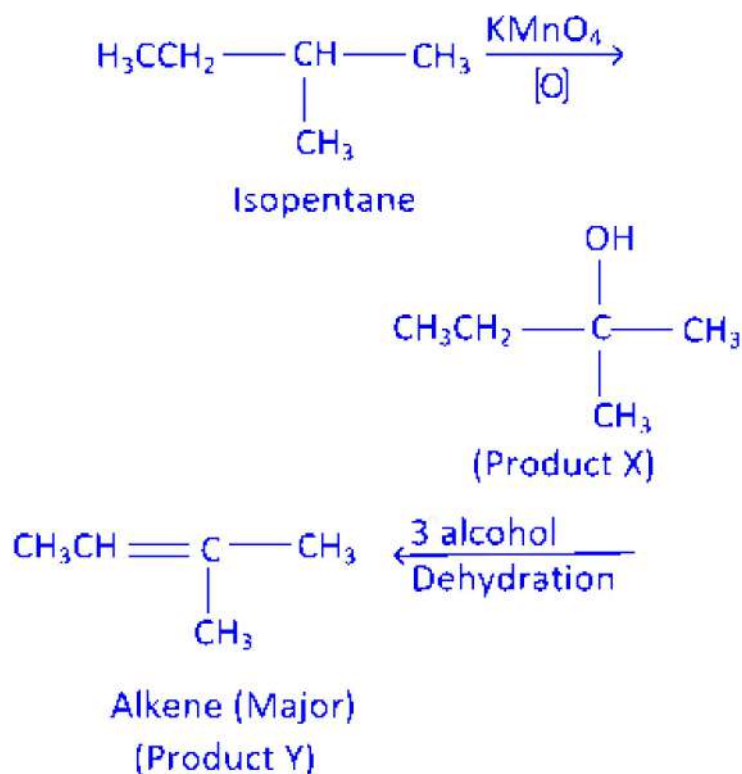
Options:

- A. 
- B. 
- C. 
- D. 

Answer: A

Solution:

The complete reaction is as follows,



Question24

The product formed when a hydrocarbon X of molecular formula C_6H_{10} is reacted with sodamide is subjected to ozonolysis, followed by hydrolysis with $\text{Zn}/\text{H}_2\text{O}_2$ and upon further oxidation gave two carboxylic acids, of which one is optically active. The hydrocarbon ' X ' is

AP EAPCET 2021 - 19th August Evening Shift

Options:

- A. hex-1-yne
- B. hex-3-yne
- C. 3-methyl pent-1-yne
- D. 3,3-dimethyl but -1-yne

Answer: C

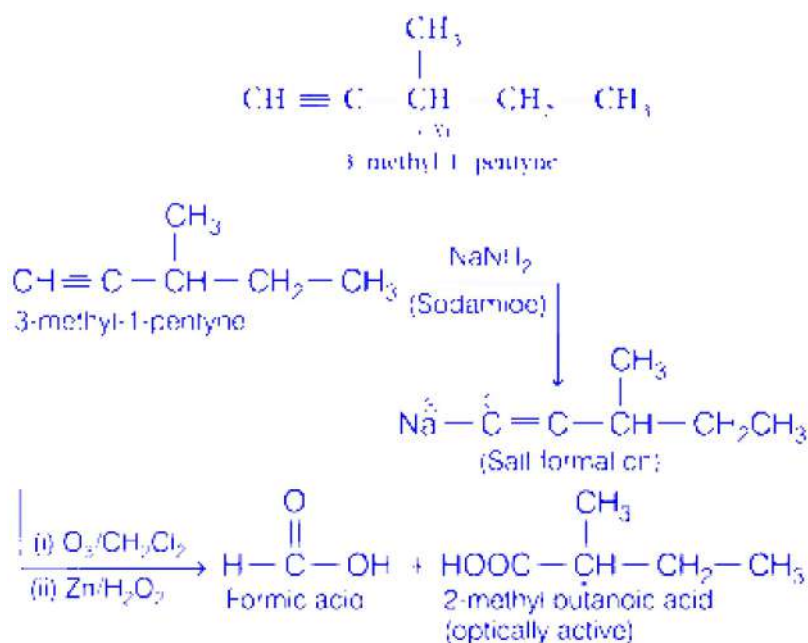
Solution:

Here for unknown formula given C_6H_{10} , we can predict molecular structure of given formula by using double bond equivalent.

$$DBE = C - \frac{H}{2} + \frac{N}{2} - \frac{X}{2} + 1$$

$$\Rightarrow = 6 - \frac{10}{2} + 0 - 0 + 1$$

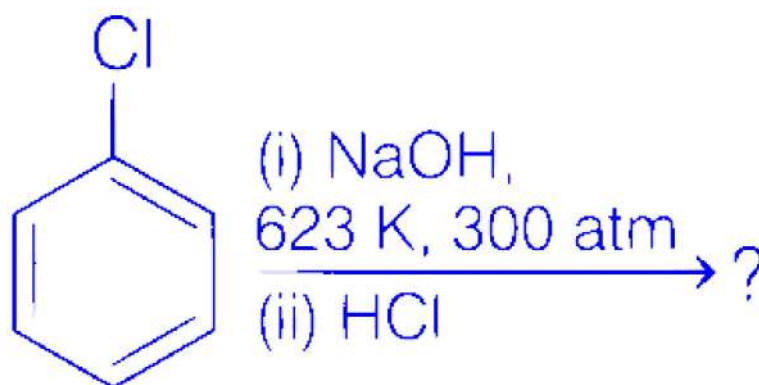
$$\Rightarrow = 6 - 5 + 1 = 2$$



Hence, the hydrocarbon X is 3-methylpent-1-yne.

Question25

Identify the product of the following reaction.



AP EAPCET 2021 - 19th August Evening Shift

Options:

- A. Benzene
- B. Toluene
- C. Aniline
- D. Phenol

Answer: D

Solution:

When chlorobenzene is heated with 6% solution of NaOH at 623 K and under pressure 300 atm to form sodium phenoxide.

Further on acidified with dilute HCl, it will form phenol.

